Article

SEMTRA: Global Semantic Transition and Rough-Set Rules for Auditable Post-hoc Explainability

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Abstract

Deep learning architectures produce effective but difficult-to-audit latent representations, creating a practical gap between predictive performance and verifiable explanation. We propose Global Semantic Transition (SEMTRA), a post-hoc framework that maps frozen representation features into semantic attributes, discretizes those attributes, and induces rough-set production rules with explicit coverage, conflict, fidelity, and abstention reporting. In five-seed AwA2 Protocol A experiments, the semantic transition achieved a test mean absolute error of 0.1029 ± 0.0005. The extracted rulebook covered 84.80% of test instances, with covered accuracy of 39.73% and covered fidelity to the base predictor of 40.48%. Under the official xlsa17 Protocol B split, continuous semantic-prototype transfer reached $44.02\% \pm 1.22\%$ unseen-object accuracy; this result is reported as semantic-transfer validation rather than as a competitive zero-shot learning claim. Cross-domain SUN and Derm7pt technical validations further show that the same audit protocol is portable but strongly dataset-dependent. In the controlled synthetic benchmark, SEMTRA achieved macro-F1 = 0.879 at zero semantic noise and 0.838 at the highest evaluated noise level; the mean across evaluated noise levels was 0.8668. These results support SEMTRA as an audit layer for exposing the verifiable subset of a model's semantic logic, not as a replacement for the underlying predictor.

Keywords: explainable artificial intelligence; transition matrix; semantic attributes; rough sets; production rules; discretization; zero-shot learning

1. Introduction

Deep learning architectures have established themselves as definitive standards for visual recognition and continuous transfer learning because of their inherent ability to compress high-dimensional observations into remarkably effective latent representations. However, this compression mechanism inherently creates a severe black-box effect, meaning that neural features remain structurally optimized for predictive success rather than for human interpretability. As automated systems increasingly permeate mission-critical diagnostic domains, the demand for transparency has fundamentally shifted away from heuristic approximations toward rigorous, mathematically sound explanations. The recent manifesto detailing the evolution of Artificial Intelligence (AI) frameworks emphasizes the critical need for verifiable symbolic knowledge [1].

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Current approaches addressing this transparency demand fall into diverse methodological categories. A comprehensive survey of modern explainable frameworks illustrates a prominent divide between local diagnostic techniques and globally interpretable model designs [2]. The problem under consideration is the inability to systematically translate compressed latent neural coordinates into a robust, global, and conflict-aware rulebook without heavily retraining or modifying the underlying feature extractor. While instance-based post-hoc explanations generate input-specific attribution maps, they completely fail to produce a holistic, reusable logic policy for rigorous auditing. Conversely, interpretable-by-design architectures necessitate predefined concept vocabularies and costly structural re-engineering of existing high-performance networks. This persistent interpretability gap severely impairs routine model verification and diagnostic safety. Consequently, it is highly important to present novel scientific contributions to address the raised problem by synthesizing a global mathematical bridge that connects uninterpretable latent manifolds with auditable symbolic spaces.

To construct this required bridge, the proposed SEMTRA framework explicitly implements a post-hoc continuous semantic transition. Early analytical studies validated the efficacy of transition matrices for interpreting complex networks within structured visual analytics environments [3]. These matrix structures were subsequently adapted and refined to extract understandable features in specialized healthcare informatics contexts [4]. Furthermore, theoretical research extended linear translation operators to manage continuous geometric symmetries via equivariant algebraic manifolds [5]. SEMTRA advances this trajectory by converting the extracted continuous semantic variables into verifiable logical predicates using established rough-set principles. Decades ago, initial mathematical formalizations introduced the concept of bounding uncertainty through precise indiscernibility relations [6]. Comprehensive theoretical extensions eventually solidified the applicability of rough sets in structured reasoning about inconsistent data [7].

The present manuscript is distinct from the prior Equivariant Transition Matrix (ETM) study. ETM targets symmetry-aware transition operators and Lie-group linearization, with the central objective of preserving geometric consistency under specified transformations [5]. SEMTRA is not an equivariance method and does not claim Lie-group structural consistency. Its contribution is the semantic-to-symbolic audit sequence: semantic reconstruction, discretization, rough-set granulation, conflict-aware rule induction, abstention, and quantitative audit-tax reporting. Table 1 summarizes this separation.

Table 1. Comparison of the prior ETM framework and the present SEMTRA framework.

Dimension	ETM paper	SEMTRA manuscript	Relationship
Core problem	Symmetry-consistent explanation mapping	Semantic auditability of frozen representations	Complementary post-hoc transition goals
Mathematical operator	Lie-group-constrained transition matrix	Ridge semantic transition plus rough-set rules	Shared transition family, different constraints
Constraint	Equivariance and Lie-algebra intertwining	Discretization stability, support, confidence, and abstention	Non-overlapping objective functions
Semantic layer	Geometric mental-model coordinates	Human-interpretable semantic attributes	SEMTRA makes semantics the rule substrate
Discretization	Not central	WEDD, entropy, quantile, and width comparisons	SEMTRA adds symbolic state construction
Symbolic rule induction	Not included	Rough-set granules, reducts, and production rules	SEMTRA adds auditable rulebooks
Conflict/abstention	Not primary output	Explicit conflict and structural abstention metrics	SEMTRA exposes unverifiable regions
Primary output	Symmetry-aware transition explainer	Coverage-aware semantic rulebook	Different audit artifacts
Evaluation focus	Geometric consistency and reconstruction	Fidelity, coverage, accuracy, conflict, runtime, and sensitivity	Distinct empirical questions

The goal of this study is to improve the auditability, transparency, and logical verifiability of pre-trained deep learning representations by applying a continuous semantic transition coupled with rigorous rough-set knowledge granulation.

The primary objective of this study is to formulate and empirically evaluate the SEMTRA framework, ensuring it effectively translates latent neural features into a conflict-aware symbolic rulebook while explicitly identifying ambiguous boundary regions. To fulfill this objective, the research proposes four major contributions:

- A dimensionally explicit transition-matrix formulation that maps compressed deep representations to interpretable semantic attributes without retraining the underlying feature extractor.
- A discretization module, instantiated by WEDD and benchmarked against entropy and quantile alternatives, for converting reconstructed semantics into symbolic states.
- A rough-set rule-induction mechanism that constructs a global, conflict-aware production rulebook, featuring explicit handling of structural abstentions and boundary regions.
- A reviewer-driven evaluation on AwA2, SUN, Derm7pt, and controlled synthetic data, demonstrating how coverage, fidelity, accuracy, conflict, sensitivity, and runtime jointly define the audit tax.

The remainder of this paper is organized as follows. Section 2 details the related works covering explainable methods and symbolic rules. Section 3 outlines the proposed framework methodology, the continuous transition bridge, and the subsequent rough-set extraction mechanisms. Section 4 presents the qualitative and quantitative evaluations. Section 5 analyzes the implications of the structural explanation tradeoff. Finally, Section 6 concludes the manuscript.

2. Related works

2.1. Local post-hoc explanation

Local post-hoc interpretation approaches estimate decision boundaries in the immediate vicinity of individual input samples to generate explanations. Highly ubiquitous frameworks, such as Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), heavily rely on perturbed neighborhood samples to construct sparse additive feature attributions [8]. Despite their widespread acceptance, these instance-specific models output fragmented attribution heatmaps rather than unified global policies. Recent advancements in local explainability address logical consistency by deploying universal rule-based explainers designed to maintain high local precision via conditional extraction paths [9]. Additionally, novel fusion methodologies combine independent feature importance rankings with structured local rule induction to stabilize the underlying attribution logic [10]. However, even optimized local methodologies fail to yield the exhaustive, dataset-wide conflict resolution protocols essential for comprehensive diagnostic auditing.

2.2. Concept-based and Ante-hoc models

Concept-based techniques aim to directly augment numerical latent features with variables mapped to human-understandable terms. The foundational introduction of concept activation vectors provided a mathematical formulation to determine whether final neural decisions remain sensitive to predefined semantic directions [11]. More sophisticated architectural approaches, specifically Concept Bottleneck Models (CBMs), require networks to accurately predict intermediate semantic variables before executing final classification outputs [12]. Contemporary literature illustrates that these discrete human-aligned concepts can be natively embedded within models during the initial structural training phase [13].

Alternatively, selected optimization algorithms can reliably infer similar aligned concepts entirely through post-hoc modifications [14]. The primary limitation of ante-hoc methodologies remains their dependence on specialized initial training regimens, hindering their straightforward application to widely deployed black-box feature extractors.

2.3. Formal Knowledge Granulation and Rough Sets

The scientific pursuit of inducing symbolic rule structures predates the massive expansion of deep artificial neural networks. A comprehensive review highlights that classic hierarchical decision trees organically partition observations based on localized entropy metrics to generate completely transparent rule sequences [15]. Concurrently, separate-and-conquer logic algorithms iterate over modular rules to balance overall predictive accuracy with ultimate structural compactness, a process recently enhanced by integrating fuzzy logic components [16]. Modern structured applications successfully deploy these distinct rule-based algorithms to dissect complex unsupervised anomaly detection frameworks [17]. Nevertheless, most heuristic rule inducers forcefully dictate deterministic boundary classifications. Formulating stable partitions for rough-set structures necessitates robust preprocessing; modern non-monotonic discretization techniques demonstrate substantial utility in stabilizing interval boundaries [18]. Furthermore, novel kernel density estimators effectively map topological probability distributions within large, highly dimensional continuous spaces [19].

2.4. Semantic Transfer as a Validation Proxy

Continuous semantic transfer evaluation serves as a robust mechanism to assess whether a given methodology successfully captures generalized conceptual structures. For instance, the AwA2 benchmark facilitates rigorous testing of continuous semantic projection between seen domains and unseen domains [20]. This foundational repository provides standardized imagery accompanied by precise numeric class-level semantic prototype vectors [21]. Historically, the Direct Attribute Prediction (DAP) mechanisms established baseline predictive benchmarks through probabilistic semantic mappings [22]. Subsequent Indirect Attribute Prediction (IAP) formulations refined structural relationships among intermediate variables to better generalize conceptual knowledge [23].

To continuously elevate zero-shot classification performance, researchers have developed embedding, generative, attention-based, and vision-language architectures. Generative Framework for Zero-Shot Learning (GFZSL) improves zero-shot accuracy by synthesizing pseudo-samples derived from attribute distributions [24], while f-VAEGAN-D2 extends feature generation to generalized any-shot settings [25]. Dense attribute attention and attribute-prototype mechanisms localize discriminative semantic evidence more explicitly [26,27], and contrastive or transformer-based systems further improve visual-semantic alignment [28–31]. Diverse analytical methodologies calculate probabilistic convex combinations utilizing fixed semantic embeddings [32], while cross-modal transfer architectures project textual descriptions into visual coordinates [33]. Structured optimization strategies apply semantic similarity embedding paradigms [34], specialized label-embedding operations [35], and structured joint formulations [36]. Other notable approaches range from fast linear regularizations [37] and synthetic class-weight hallucinations [38] to autoencoder architectures that preserve topology within projection domains [39]. Because these spaces remain high dimensional, robust algorithmic implementations rely on efficient randomized SVD matrix decompositions [40].

Despite high accuracy, current zero-shot learning methods lack transparency. They rely on complex, non-linear models that act as secondary black boxes, completely hiding how visual features map to semantic concepts. By prioritizing predictive performance, they sacrifice mathematical traceability and make it impossible to track exact decision pathways. In contrast, the proposed framework treats semantic transfer strictly as a validation tool rather than a predictive competition. It employs a transparent transition mechanism to ensure all recovered concepts can be rigorously verified, avoiding the opacity of standard embedding techniques.

Broadly, the Explainable Artificial Intelligence (XAI) landscape faces several critical challenges: local methods lack global consistency, interpretable-by-design models require expensive retraining, and traditional rule learners force decisions even when data is ambiguous. SEMTRA addresses these gaps by combining a global, explicitly defined transition matrix with rough-set granulation. By prioritizing mathematical traceability and conflict awareness over raw predictive optimization, SEMTRA provides a verified post-hoc framework capable of extracting honest, reproducible diagnostic rules from pre-trained representations.

3. Materials and methods

3.1. Framework Architecture Overview

The SEMTRA framework is structurally designed as a modular, post-hoc audit pipeline for settings where frozen representation features can be aligned to a semantic attribute dictionary. Operating over a frozen feature extractor, the methodology produces a global symbolic rulebook for the subset of model behavior that can be reconstructed, discretized, and verified. The conceptual architecture, illustrated in Figure 1, explicitly follows a two-phase progression: Phase 1, designated as Neural-to-Semantic Transition, and Phase 2, defined as Semantic-to-Symbolic Granulation.

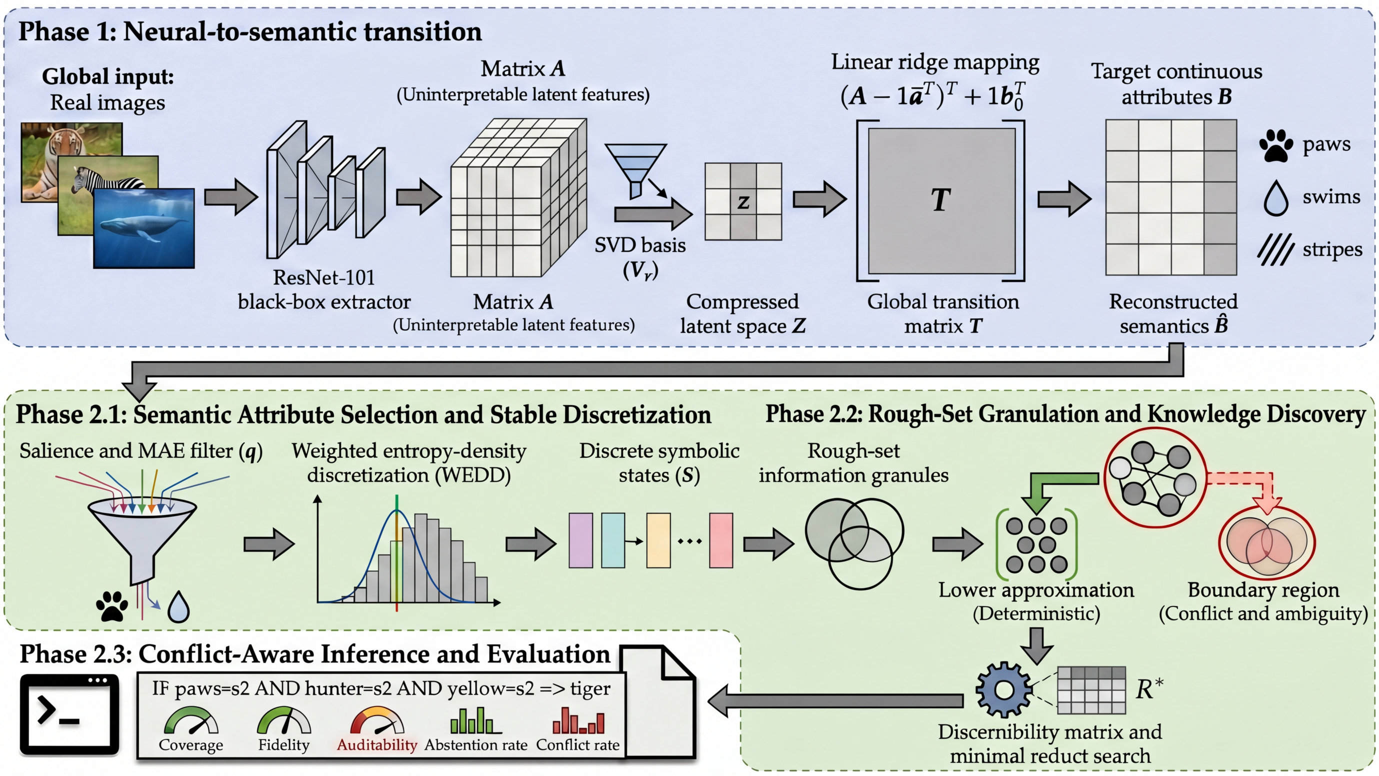


Figure 1. Conceptual architecture of the SEMTRA framework. The end-to-end pipeline sequentially translates uninterpretable latent neural features into continuous semantic attributes via a global transition matrix, and subsequently leverages density-aware discretization and rough-set granulation to induce a transparent, conflict-aware rulebook.

The progression moves from the initial latent space $\mathbf{A}$, mapped through a robust global transition matrix $\mathbf{T}$, to reconstruct understandable continuous semantics $\mathbf{B}$. Subsequently, these semantics are discretized through the WEDD mechanism to finally induce the optimal symbolic production rules $\mathcal{R}$. To ensure absolute notational clarity and resolve discrete representation overlaps, the matrix of all discretized states is strictly denoted as $\mathbf{S}$, with individual symbolic states for object i and attribute j defined as s_{ij}. The ultimate output of the framework structurally follows the standardized rule format r:

$$r: \quad \bigwedge_{j \in R_r} [s_j = v_j] \Rightarrow d = y_r, \tag{1}$$

where R_r is the optimized minimal set of semantic antecedents, s_j represents the specific discrete state variable required by the rule, and v_j is the corresponding exact logical value bound.

The mathematical formalization of the induced decision-making system S is defined as:

$$S = (\mathcal{U}, \mathcal{C}, d, \mathcal{V}, f), \tag{2}$$

where $\mathcal{U}$ is the universe of distinct objects, $\mathcal{C}$ is the set of conditional semantic attributes, d is the target decision attribute, $\mathcal{V}$ is the union of valid attribute domains, and f is the deterministic information function.

By explicitly separating heavy computational inference from lightweight symbolic extraction, SEMTRA makes the retained explanations traceable and reproducible under the stated semantic-attribute and split assumptions.

3.2. Phase 1: Neural-to-Semantic Transition

The first phase establishes a mathematically explicit bridge between the uninterpretable latent representation space of a deep model and a target semantic attribute space. Let $\mathbf{A} \in \mathbb{R}^{m \times k}$ denote the matrix of raw, untransparent latent features. To ensure that the subsequent mapping is robust to dataset-specific biases, we first perform dimensional centering. Let the vector $\bar{\mathbf{a}} \in \mathbb{R}^k$ denote the absolute mean computed across the entire training distribution. The centered representation matrix $\mathbf{A}_c$ is subsequently computed via:

$$\mathbf{A}_c = \mathbf{A} - \mathbf{1}\bar{\mathbf{a}}^\top \in \mathbb{R}^{m \times k}. \tag{3}$$

To handle potential noise and extreme collinearity within high-dimensional latent spaces commonly found in complex architectures, a rank-r SVD basis, denoted as $\mathbf{V}_r \in \mathbb{R}^{k \times r}$, is rigorously extracted [40]. This yields a computationally compressed latent space $\mathbf{Z}$:

$$\mathbf{Z} = \mathbf{A}_c \mathbf{V}_r \in \mathbb{R}^{m \times r}. \tag{4}$$

The mapping to a human-understandable target semantics matrix $\mathbf{B} \in \mathbb{R}^{m \times \ell}$, where ℓ represents the count of conceptually interpretable variables, is established via a ridge-regression weight matrix $\mathbf{W} \in \mathbb{R}^{r \times \ell}$, formalized as the following minimization problem:

$$\mathbf{W}^* = \arg\min_{\mathbf{W} \in \mathbb{R}^{r \times \ell}} \|\mathbf{Z}\mathbf{W} - \mathbf{B}\|_F^2 + \alpha \|\mathbf{W}\|_F^2. \tag{5}$$

The choice of a linear ridge operator is an intentional methodological decision. We deliberately employ a linear transition matrix because it is interpretable in itself; human auditors can directly inspect the weights of each feature to track structural contributions. Furthermore, linearity avoids the opaque boundary distortions and gradients typically associated with deep non-linear alternatives. With an estimated analytical intercept $\mathbf{b}_0 \in \mathbb{R}^\ell$, the primary global transition operator $\mathbf{T}$ and the reconstructed continuous attributes $\widehat{\mathbf{B}}$ are recovered:

$$\mathbf{T} = \mathbf{V}_r\mathbf{W}^* \in \mathbb{R}^{k\times\ell}, \quad \widehat{\mathbf{B}} = (\mathbf{A} - \mathbf{1}\bar{\mathbf{a}}^\top)\mathbf{T} + \mathbf{1}\mathbf{b}_0^\top. \tag{6}$$

This linear transformation enforces the model's internal logic into a semantically grounded coordinate system. Global semantic reconstruction fidelity is strictly evaluated using the Mean Absolute Error (MAE):

$$\mathrm{MAE}_Q = \frac{1}{|Q|\ell}\sum_{i\in Q}\sum_{j=1}^{\ell}|\hat{b}_{ij} - b_{ij}|. \tag{7}$$

3.3. Phase 2.1: Semantic Attribute Selection and Stable Discretization (WEDD)

Once the continuous semantic attributes $\widehat{\mathbf{B}}$ are reconstructed, the next operation isolates a highly relevant subset of q predictive attributes (default $q = 18$ in the v1 experiments) to prevent computational combinatorial explosion. Let e_j signify the normalized validation MAE for attribute j, and let $\eta_j = \|\mathbf{T}_{:j}\|_2$ represent the mathematical transition salience. The formal selection score is defined as:

$$\mathrm{score}_j = \lambda_s\widetilde{\eta}_j + (1-\lambda_s)(1-\widetilde{e}_j). \tag{8}$$

where $\widetilde{(\cdot)}$ explicitly denotes Min-Max normalization mapping values robustly into the $[0,1]$ interval across all ℓ available target attributes.

The parameters λ_s and λ_H serve as dedicated auditor's control knobs. Specifically, $\lambda_s \in [0,1]$ allows the user to mathematically prioritize either the strength of the neural-to-semantic signal (purity) or the precision of the reconstruction error. A comprehensive sensitivity analysis justifying the default configuration of the auditor's control knobs ($\lambda_s = 0.5, \lambda_H = 0.65$) is provided in Appendix A, demonstrating their direct impact on rulebook compactness and coverage. The selected attributes are subsequently transformed into discrete symbolic intervals using the WEDD mechanism.

For a candidate boundary threshold θ splitting an attribute x_j into a left computational subset L_θ and a corresponding right subset R_θ, the standard conditional decision entropy $H_j(\theta)$ is calculated using Equation (9) and Equation (10):

$$H_j(\theta) = \frac{|L_\theta|}{n}H(d \mid L_\theta) + \frac{|R_\theta|}{n}H(d \mid R_\theta), \tag{9}$$

$$H(d \mid G) = -\sum_{y=1}^{N_c} p(y \mid G)\log_{N_c} p(y \mid G), \tag{10}$$

where N_c is the number of distinct decision classes, ensuring that $H(d|G) \in [0,1]$ perfectly across all evaluated datasets, thereby stabilizing the final WEDD objective across vastly different problem scales.

To prevent jitter in the resulting logic, WEDD introduces a local probability density proxy $p_j(\theta)$, computed via a Gaussian kernel function $K(u)$ with a structural bandwidth h_j:

$$p_j(\theta) = \frac{1}{nh_j}\sum_{i=1}^{n} K\left(\frac{\theta - x_{ij}}{h_j}\right). \tag{11}$$

The final discretization threshold is determined by minimizing the unified WEDD multiobjective function:

$$J_j(\theta) = \lambda_H\widetilde{H}_j(\theta) + (1-\lambda_H)\widetilde{p}_j(\theta). \tag{12}$$

Explicitly, both $\widetilde{H}_j(\theta)$ and $\widetilde{p}_j(\theta)$ are precisely scaled utilizing standard Min-Max normalization into the $[0, 1]$ interval across all generated candidate thresholds, guaranteeing the multiobjective weights behave uniformly and symmetrically.

WEDD does not simply minimize entropy as standard decision trees do. The parameter $\lambda_H \in [0, 1]$ is an auditor-facing knob that trades rule informational purity against density-aware boundary placement. Because the v1 ablation shows that MDLP-like entropy can match or exceed WEDD on several AwA2 audit metrics, WEDD is treated here as a stability-oriented design option rather than as an unconditionally superior discretizer. The complete procedural implementation of these early phases is outlined in Algorithm 1.

Algorithm 1 Neural-to-Semantic Transition and WEDD Discretization (Phases 1 and 2.1)

```
Input: Universe of uninterpretable representations A, aligned continuous target attributes B, decision labels d.
Output: Discretized symbolic states S.
Initialize the global decision-making system S via Equation (2).
Compute the centered representation matrix A_c utilizing Equation (3).
Extract the rank-reduced compressed latent space Z via Equation (4).
Optimize the ridge-regression weight matrix W* applying Equation (5).
Reconstruct continuous semantic attributes B̂ and establish the global transition operator T via Equation (6).
Evaluate the semantic reconstruction fidelity by computing the MAE via Equation (7).
Compute the composite attribute selection score score_j using Equation (8) and isolate q highly relevant concepts.
for each selected attribute j ∈ {1, ..., q} do
    Compute the conditional decision entropy H_j(θ) for candidate thresholds utilizing Equation (9) and Equation (10).
    Estimate the local probability density proxy p_j(θ) applying Equation (11).
    Determine the optimal discretization threshold by minimizing the unified WEDD objective J_j(θ) via Equation (12).
end for
Map the continuous attributes into the discrete symbolic states S.
Return S
```

3.4. Phase 2.2: Rough-Set Granulation and Knowledge Discovery

Following stable discretization, the framework constructs a symbolic logic layer through mathematically formal knowledge granulation. Formally, the support of a specific rule r ($A \Rightarrow B$) is defined as the absolute number of instances satisfying both the antecedent and the consequent simultaneously: $\mathrm{Supp}(r) = |A \cap B|$. The confidence is subsequently formulated as the conditional probability of the consequent given the antecedent: $\mathrm{Conf}(r) = \frac{|A \cap B|}{|A|}$. The core of this robust process is the establishment of a rigorous indiscernibility relation. Two distinct objects x_i and x_j are rendered indiscernible with respect to a subset of optimal attributes $P \subseteq \mathcal{C}$ if they logically share perfectly identical discrete symbolic signatures. Formulated in Equation (13), this relation partitions the universe of objects into tightly bound structural information granules, denoted as $[x]_P$:

$$IND(P) = \{(x_i, x_j) \in \mathcal{U} \times \mathcal{U} \mid \forall c_k \in P, f(x_i, c_k) = f(x_j, c_k)\}, \tag{13}$$

where $f(x_i, c_k) = s_{ik}$ explicitly utilizes the information function f from the defined decision system to represent the discrete symbolic state mapping of object x_i evaluated across the specific attribute c_k.

This precise granulation allows the framework to explicitly distinguish between deterministic logical certainties and structural ambiguities. For a specific analytical decision class Y, we systematically define the structural Lower Approximation $\underline{P}Y$:

$$\underline{P}Y = \{x \in \mathcal{U} \mid [x]_P \subseteq Y\}. \tag{14}$$

The Lower Approximation encapsulates all information granules that completely and unambiguously belong to class Y, forming the irrefutable foundation for generating perfectly certain rules. Conversely, granules that simultaneously intersect multiple diverse classes form the Boundary Region. This Boundary Region mathematically identifies the exact conflict zones where the compressed latent neural features become inherently indiscernible and fundamentally unreliable.

To ensure that the resulting explanations are conceptually compact, we sequentially apply a mathematically rigorous minimal reduct search. A reduct R^* is the smallest possible subset of attributes that preserves decision-making consistency. The generalized minimal reduct R^*_ρ optimization is formally defined as:

$$R^*_\rho = \arg\min_{R \subseteq \mathcal{C}_\rho} |R| \quad \text{s.t.} \quad \mathrm{Conf}(R \Rightarrow d_\rho) \geq \tau, \quad \mathrm{Supp}(R \Rightarrow d_\rho) \geq s_{\min}, \tag{15}$$

where ρ fundamentally denotes a specific target decision class.

To efficiently navigate this logic, the framework formulates a global discernibility matrix. Each interactive element c_{ij} securely identifies the exact set of differentiating semantic attributes between two differently classified objects:

$$c_{ij} = \{c_k \in \mathcal{C} \mid s_{ik} \neq s_{jk}\}, \quad \text{for } y_i \neq y_j. \tag{16}$$

3.5. Phase 2.3: Conflict-Aware Inference and Evaluation

The final phase of the SEMTRA framework governs the application of the induced rulebook to unobserved data. During real-time analytical inference, new object representations flow through the identical transition and discretization sequence established in Phases 1 and 2.1, instantly converting raw neural features into active symbolic signatures. This ensures that the explanation logic remains strictly faithful to the original model's "view" of the data.

3.5.1. Conflict Resolution and Soft Matching

When querying the generalized rulebook, two types of logical edge cases may emerge: boundary conflicts and structural gaps. Boundary conflicts occur when multiple activated rules suggest different outcome classes for the same symbolic signature. These are systematically resolved using a carefully constructed integral support score $S(y_k)$, as formulated in Equation (17):

$$S(y_k) = \sum_{r \in \mathcal{R}_{\text{active}}, d_r = y_k} \mathrm{Conf}(r) \cdot \mathrm{Supp}(r) \cdot w_r, \tag{17}$$

where $w_r = \left(\frac{1}{|R_r|} \sum_{j \in R_r} p_j(\theta_j)\right)^{-1}$ functions as a targeted density-based weight strictly inherited from the WEDD discretization phase, explicitly taking the inverse of the average probability density evaluated precisely at the thresholds utilized in rule r, effectively prioritizing rules fundamentally anchored in stable, structurally optimal attribute regions.

In scenarios where no exact production rules match the presented inference data (structural gaps), the framework employs soft analytical matching. This fallback mechanism evaluates partial logical alignments using the fundamental symbolic Hamming distance d_H:

$$d_H(\mathbf{s}_i, \mathbf{t}_y) = \frac{1}{|R_y|} \sum_{j \in R_y} \mathbb{I}[s_{ij} \neq t_{yj}], \tag{18}$$

where $\mathbf{s}_i$ is the signature of the test instance and $\mathbf{t}_y$ is the formal rule antecedent.

This soft matching natively utilizes a masked Hamming distance evaluated precisely only over the highly specific antecedent conditions $|R_y|$ encoded within rule $\mathbf{t}_y$. If the computed distance formally exceeds a predefined tolerance threshold ($\tau_H = 0.25$), the overall system correctly triggers a structural abstention, firmly identifying the evaluated case as currently unverifiable within the generated rulebook.

3.5.2. Fidelity and Coverage Metrics

To quantify the quality of the extracted explanation layer, SEMTRA distinguishes between two levels of fidelity. First, Covered Fidelity evaluates the accuracy of the rulebook exclusively on instances where a definitive rule was matched:

$$\text{Fidelity}_{\text{covered}} = \frac{1}{|C_Q|} \sum_{i \in C_Q} \mathbb{I}[r(u_i) = g(u_i)], \tag{19}$$

where $g(u_i)$ is the prediction of the base black-box model and C_Q is the deterministically covered subset.

Second, All-Object Fidelity logically treats required abstentions as explicit structural failures, providing a conservative measure of the framework's total explanatory power:

$$\text{Fidelity}_{\text{all}} = \frac{1}{|Q|} \sum_{i \in Q} \mathbb{I}[r(u_i) = g(u_i)]. \tag{20}$$

This dual-metric approach formalized by Equations (19) and (20) allows auditors to assess the important tradeoff between explanation precision and rulebook coverage.

Finally, the entire sequential progression, from granulation to conclusive inference, is synthesized in Algorithm 2.

3.6. Evaluation Metrics for Auditability

To evaluate SEMTRA as an XAI proof-of-concept, we transcend standard predictive accuracy with a multi-dimensional suite of metrics focused on auditability, quantifying the trade-off between extracted logic complexity and faithfulness to the deep model. Rulebook Coverage (Cov) represents explanation "breadth," measuring the percentage of tested instances Q receiving a deterministic symbolic explanation:

$$\text{Cov} = \frac{|C_Q|}{|Q|}. \tag{21}$$

High coverage denotes a rulebook that captures a large portion of the evaluated space. Covered Fidelity (F_{cov}) measures agreement between the rulebook prediction $\hat{y}_{\mathcal{R}}(x_i)$ and the frozen base-model prediction $f_{\text{BB}}(x_i)$ on the covered subset, whereas Covered Accuracy measures agreement between $\hat{y}_{\mathcal{R}}(x_i)$ and the ground-truth label y_i on that same subset. All-object accuracy counts abstentions as failures against y_i, and all-object fidelity counts abstentions as failures against $f_{\text{BB}}(x_i)$. The Structural Abstention Rate (Abs) identifies regions where the neural representation is too ambiguous or conflicted for rules:

$$\text{Abs} = 1 - \text{Cov}. \tag{22}$$

Algorithm 2 Rough-Set Granulation, Reduct Solver, and Inference (Phases 2.3–2.5)

Input: Discretized symbolic states $\mathbf{S}$, decision vector $\mathbf{d}$, validation instances.
Output: Conflict-aware production rulebook $\mathcal{R}$ and inference predictions.
Establish the indiscernibility equivalence relations $IND(P)$ across all discrete states utilizing Equation (13).
Aggregate objects into information granules and isolate the deterministic boundary using the lower approximation $\underline{P}Y$ via Equation (14).
Formulate the comprehensive discernibility matrix M, populating individual elements c_{ij} using Equation (16).
Execute the weighted heuristic search to extract the minimal optimal reducts R_{ρ}^{*} applying Equation (15).
Synthesize the discrete granules and reducts into the formal logical rule format r via Equation (1).
Compile the verified rules into the global conflict-aware rulebook $\mathcal{R}$.
for each unseen test instance **do**
    Attempt to match the instance against the rulebook antecedents.
    if multiple conflicting rules are activated **then**
        Resolve the boundary conflict by maximizing the integral support score $S(y_k)$ utilizing Equation (17).
    else if no exact rule conditions are satisfied **then**
        Execute soft matching by minimizing the symbolic Hamming distance d_H via Equation (18).
    end if
end for
Assess the overarching algorithmic explanation quality by calculating covered fidelity via Equation (19) and all-object fidelity via Equation (20).
Return $\mathcal{R}$

Rulebook Compactness evaluates the total number of rules and the average number of antecedents per rule; following the principle of cognitive load, a smaller, shallower rulebook is inherently more auditable by human experts. Finally, the Conflict Rate (CR) tracks boundary conflicts during inference, diagnosing inherent "indiscernibility" in underlying latent representations. Calculated from instances X_{conflict} mapping directly into conflict zones:

$$\mathrm{CR} = \frac{|X_{\text{conflict}}|}{|Q|}. \tag{23}$$

This metric suite is target-explicit: semantic reconstruction targets $\mathbf{B}$, fidelity targets the frozen predictor, and accuracy targets dataset labels.

3.7. Experimental Protocol and Proof-of-Concept Design

To validate the SEMTRA framework as a reliable instrument for post-hoc deep learning auditing, a multi-stage experimental protocol evaluates how it bridges continuous latent representation spaces and discrete logic, manages structural uncertainty, and yields highly auditable verification artifacts across real-world semantic reconstruction, zero-shot logical transfer, algorithm stability, and synthetic baseline recovery.

3.7.1. Datasets, Feature Extraction, and Base Modeling

Primary real-world evaluations utilize the AwA2 [21] benchmark. Eliminating image-level augmentations, the framework uses a 2048-dimensional ResNet-101 representation layer released with the xlsa17 benchmark resources [20]. A base ridge classifier is trained on the compressed representations to establish the frozen predictive target for fidelity. The v1 revision additionally evaluates SUN Attribute Database scenes [41,42] and Derm7pt / Seven-Point Checklist dermatology cases [43–45]. SUN uses the xlsa17 ResNet-101 split with 102 scene attributes. Derm7pt uses dermoscopic images, official train/validation/test case indices, grouped diagnosis labels, and the seven checklist concepts. Derm7pt results are retrospective technical-validation results only; they are not clinical deployment evidence.

3.7.2. Empirical Evaluation Splits

To isolate memorization capacity from semantic generalization, two AwA2 testing splits are used. Protocol A (closed-world fidelity) assesses neural-to-semantic reconstruction on 50 classes using MAE, RMSE, and semantic correlation, while quantifying symbolic rulebook coverage, covered fidelity, covered accuracy, all-object accuracy, conflict, and abstention. Protocol B uses the official xlsa17 split as a semantic-transfer validation proxy. For Protocol B, the transition matrix is trained exclusively on the 40 seen classes and evaluated on 10 unseen classes. These values test whether reconstructed semantics transfer across class boundaries; they are not reported as a competitive zero-shot learning leaderboard result.

3.7.3. Methodological Ablations and XAI Baselines

To contextualize scientific tradeoffs in global explainability, Transition Operator Ablation benchmarks the global linear ridge-regression operator against non-linear alternatives. Discretization Benchmarking compares WEDD against MDLP-like entropy, equal-frequency, and equal-width discretizers in the main Results section because this ablation directly changes the WEDD claim. Sensitivity experiments vary selected-attribute count q, SVD rank r, and rule confidence thresholds. Runtime instrumentation logs per-phase execution time for the transition, discretization, rule induction, and cross-domain validation steps. Explainability paradigms contrast the rough-set rulebook with heuristic symbolic models and local post-hoc attribution surrogates.

3.7.4. Algorithmic Recovery on Synthetic Data

To isolate core mathematical machinery from the inherent indiscernibility of real-world latent manifolds, a fully controlled synthetic benchmark simulates a 10-dimensional continuous feature space governed by a predefined, ground-truth logical rule dictionary. Semantic noise ($\sigma \in [0, 0.20]$) injected across random computational seeds maps the structural limits of WEDD and rough-set induction mechanisms, measuring exact threshold recovery error, rule Jaccard similarity, macro-F1 scores, and operational coverage.

3.7.5. Diagnostic Tracking and Inference Stability

A rigorous suite of visual and statistical diagnostics clarifies the proposed algorithms, assessing the balance between logical determinism and coverage via fidelity-coverage tradeoff projections. At the attribute level, target transition salience is mapped against test reconstruction errors to justify semantic selection, differentiating robust physical descriptors from high-variance semantic concepts. Real-time analytical inference is audited by generating precise rule traces mapping unseen objects to active logical antecedents, confidence parameters, and source decisions. The structural robustness of inference pathways is systematically evaluated against semantic-representation perturbations; resultant decision consistency, coverage deviation, and rule matching stability metrics are cataloged in Appendix C.

4. Results

This section presents the empirical validation of the SEMTRA framework according to the Pillars of Auditability defined in the experimental protocol. We assess the framework's ability to reconstruct semantics, generate auditable rules, and handle zero-shot transfer as a proxy for conceptual integrity.

4.1. Semantic Reconstruction Performance (Protocol A)

The first stage of the experimental validation verifies the integrity of the Neural-to-Semantic Bridge. This is a prerequisite for a reliable audit: the extracted rules can only be trusted if the underlying transition matrix successfully recovers the intended semantic concepts from the latent neural representations.

The performance of the linear transition operator under Protocol A is summarized in Table 2. On the test set, the v1 regeneration achieved an MAE of 0.1029 ± 0.0005 and a mean semantic correlation of 0.8070 ± 0.0018 across five seeds. These results indicate that the global transition matrix consistently aligns the compressed representation with the target semantic descriptors under the released AwA2 feature protocol.

Table 2. AwA2 Protocol A transition reconstruction metrics. Metrics are reported as mean ± standard deviation across five v1 seeds.

Split	MAE	RMSE	Mean semantic correlation
train	0.1027 ± 0.0003	0.1482 ± 0.0004	0.8079 ± 0.0009
validation	0.1030 ± 0.0003	0.1486 ± 0.0005	0.8065 ± 0.0014
test	0.1029 ± 0.0005	0.1484 ± 0.0007	0.8070 ± 0.0018

To justify the choice of a linear operator over more complex alternatives, an ablation study was performed (Table 3). While non-linear regressors (e.g., two-layer MLP) marginally improve reconstruction in some metrics, they introduce significant black-box opacity into the transition itself. The linear ridge transition maintains a high correlation (0.7759) and significantly higher rule coverage (0.8718 ± 0.0075) compared to non-linear variants, establishing it as the most defensible choice for an auditable framework. The near-zero rule accuracy (0.0027) for the RBF kernel ridge indicates that severe overfitting in the high-dimensional latent space collapsed the global semantic mapping entirely, underscoring the fragility of unregularized non-linear projections.

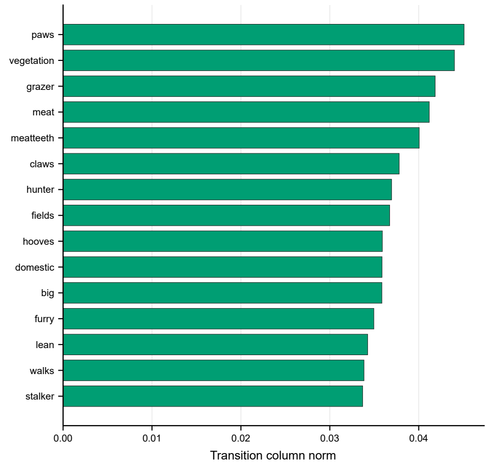
Table 3. Semantic reconstruction and rule-transfer ablation for linear and nonlinear operators. Means ± standard deviations across 5 seeds are provided for the proposed linear operator.

Operator	MAE	RMSE	Corr.	Rule cov.	Rule acc.	Runtime (s)
Linear ridge transition	0.1076 ± 0.0012	0.1573	0.7759	0.8718 ± 0.0075	0.1941	0.0451
RBF kernel ridge	0.1958	0.2583	0.1176	1.0000	0.0027	2.8573
Two-layer MLP regressor	0.1685	0.2355	0.4497	0.8390	0.0915	0.6914

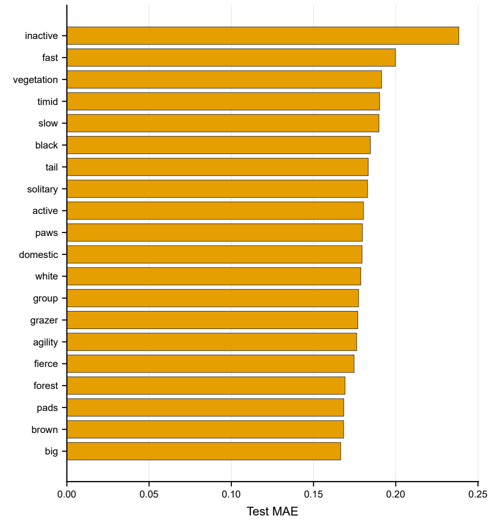
Note: All runtime measurements reported in this table were executed sequentially on a single NVIDIA RTX 3090 Graphics Processing Unit (GPU) and an Intel Core i9-10900K Central Processing Unit (CPU) to ensure a standardized computational baseline.

Detailed attribute-wise diagnostics, visualized in Figure 2, reveal that visually distinct attributes such as paws, vegetation, and water achieve the highest transition salience and lowest reconstruction errors. Conversely, more abstract descriptors (e.g., inactive) exhibit higher variance. This variance is explicitly managed in the next phase by the WEDD algorithm, which prioritizes stable, low-error attributes for rule induction.

As shown in Table 4, the composite selection score effectively isolates the top q attributes that offer the best balance between neural signal strength and reconstruction precision. For brevity, the top 12 selected attributes are shown in Table 4.



(**a**)



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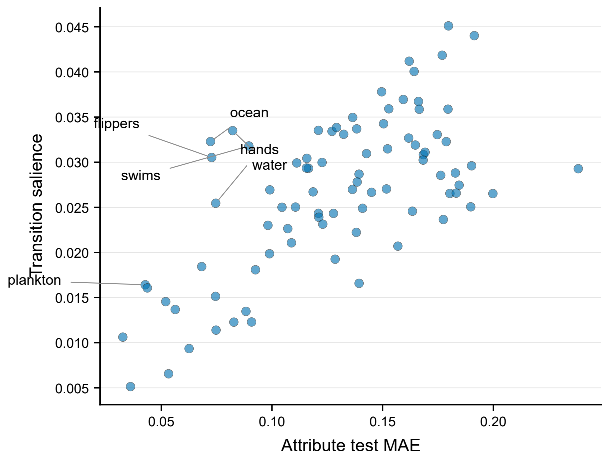


(**c**)

Figure 2. Semantic attribute diagnostics: (**a**) transition salience by semantic attribute; (**b**) attribute-wise semantic reconstruction error on AwA2; and (**c**) transition salience against test reconstruction error; labeled points are high-score selected attributes.

Table 4. Selected semantic attributes used for rough-set rule induction (first 12 shown).

Semantic attribute	Salience	Test MAE	Selection score
stripes	0.025	0.105	0.202
hooves	0.036	0.153	0.208
swims	0.032	0.090	0.288
paws	0.045	0.180	0.228
longneck	0.020	0.099	0.166
hunter	0.037	0.159	0.204
ocean	0.032	0.072	0.346
quadrapedal	0.030	0.111	0.225
water	0.033	0.082	0.325
flippers	0.031	0.073	0.323
hands	0.025	0.075	0.271
plankton	0.016	0.043	0.257

4.2. Rulebook Auditability and Fidelity

The core objective of the SEMTRA framework is to transform the continuous semantic bridge into a globally consistent, symbolic rulebook that a human auditor can inspect and verify. The properties of the generated rulebook under Protocol A are summarized in Table 5, providing a comprehensive view of the framework's explanatory power.

Table 5. General rulebook properties and symbolic baseline comparison. Point estimates are reported alongside standard deviations evaluated over 5 random seeds for the proposed method.

Metric	Value (Ours)	CART baseline	Separate-and-conquer
Number of rules	59.00 ± 2.55	53	197
Test coverage	0.8480 ± 0.0530	1.0000	0.2620
Covered accuracy	0.3973 ± 0.0109	0.4129	0.6334
Covered fidelity to base	0.4048 ± 0.0078	–	–
Average conditions per rule	4.0397 ± 0.0543	7.0000	3.2589
Test abstention rate	0.1520 ± 0.0530	0.0000	0.7380
Test conflict rate	0.1375 ± 0.0532	0.0000	0.2301

4.2.1. The Fidelity-Coverage Trade-off

The extracted rulebook achieved a mean Cov of 84.80%, while the separate-and-conquer rule learner covered only 26.20% of testing instances in the existing symbolic baseline. Its Covered Fidelity (F_{cov}) was 0.4048 ± 0.0078, meaning that the audited rules reproduced the base model's decision on approximately 40% of covered cases while still retaining explicit conflict and abstention accounting.

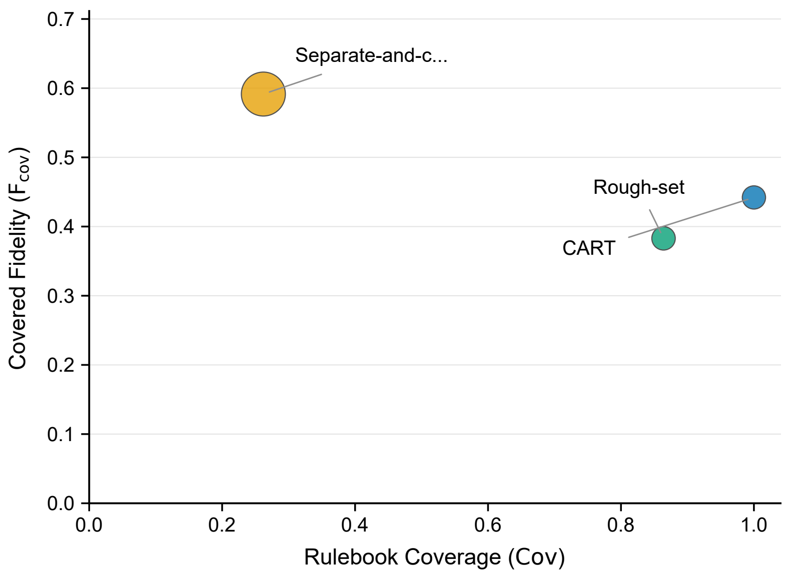


Figure 3. Symbolic baseline tradeoff between Cov and Covered Fidelity (F_{cov}). Marker size is proportional to rule count. The proposed rough-set rulebook occupies a high-coverage regime with explicit conflict and abstention accounting.

As visualized in Figure 3, SEMTRA maintains high Cov while explicitly accounting for structural uncertainty. The reduction in covered accuracy compared with the base model is a transparent audit tax–the cost of enforcing strict discretization and logical consistency on a high-dimensional non-linear system. For an auditor, the 39.73% covered accuracy and 40.48% Covered Fidelity jointly quantify the portion of the model's logic that is currently verifiable and reproducible through simple semantic predicates.

Table 6. AwA2 discretizer comparison using the same reconstructed semantics and rough-set inference logic. Values are means across seeds.

Method	Cov.	Fid.	Cov. acc.	All acc.	Conflict	Rules
Equal frequency	0.7212	0.3087	0.2999	0.2161	0.2773	50.6
Equal width	0.6401	0.4535	0.4532	0.2883	0.3599	66.8
MDLP-like entropy	0.8334	0.3975	0.3912	0.3261	0.1482	56.0
WEDD	0.8480	0.4048	0.3973	0.3373	0.1375	59.0

Table 6 moves the WEDD-vs-MDLP-like entropy ablation into the main Results section. In the v1 five-seed regeneration, WEDD has slightly higher mean coverage and covered fidelity than MDLP-like entropy, but the paired effect sizes are small. Therefore, WEDD should be interpreted as a density-aware, stability-oriented discretization option rather than as a universally superior discretizer.

Table 7. Selected-attribute-count sensitivity on AwA2 for the WEDD rulebook.

q	Coverage	Covered fidelity	Covered accuracy	Conflict	Rules
8	0.7999	0.3435	0.3422	0.1711	59
12	0.8251	0.4002	0.3889	0.1522	60
18	0.8469	0.4109	0.4054	0.1378	57
24	0.8354	0.4343	0.4266	0.1365	54
32	0.8719	0.5522	0.5380	0.1027	50

Table 8. SVD-rank sensitivity on AwA2 for semantic reconstruction and rulebook audit metrics.

Rank	MAE	Coverage	Covered fidelity	Covered accuracy	Runtime s
16	0.1295	0.9037	0.3875	0.3913	7.4
32	0.1150	0.9023	0.3668	0.3695	7.2
64	0.1031	0.8469	0.4109	0.4054	6.8
128	0.0942	0.7893	0.4898	0.4801	6.8

4.2.2. Conflict Awareness and Representative Logic

A key advantage of the rough-set approach is the explicit management of boundary regions. The framework identified a CR of 13.54%, where the neural representations of different classes became semantically indiscernible. Unlike the CART baseline, which forces a potentially misleading decision in these regions, SEMTRA triggers a structural abstention, signaling to the auditor that the model's logic is currently ambiguous for these instances.

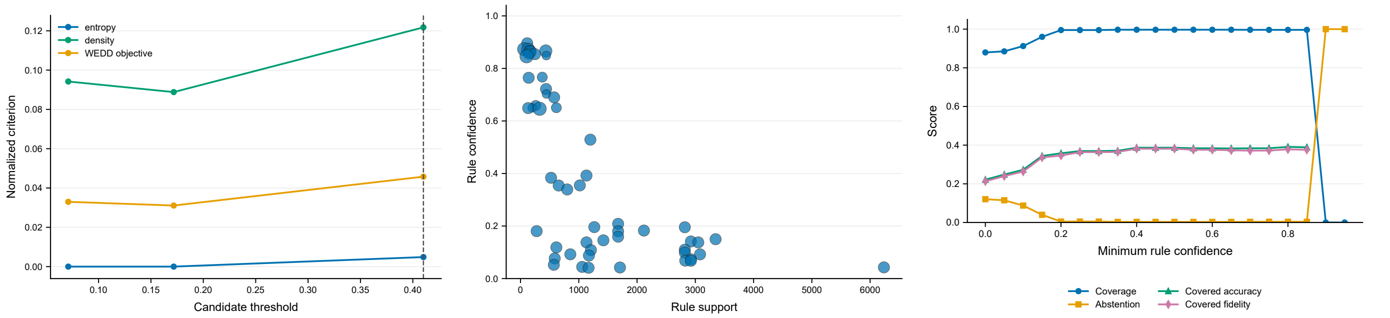
The look and feel of the resulting audit is demonstrated in Table 9, which presents representative induced rules. For example, the rule for the class tiger (R0001) relies on a stable combination of paws, hunter, and yellow attributes with a high confidence of 0.896.

Table 9. Representative induced production rules from AwA2 Protocol A.

Rule	Antecedent	Numeric bounds	Class	Support*	Conf.
R0001	paws=s2 AND hunter=s2 AND small=s0 AND yellow=s2	paws ≥ 0.40; hunter ≥ 0.40; small < 0.15; yellow ≥ 0.40	tiger	115	0.896
R0003	stripes=s1 AND hooves=s3 AND paws=s0 AND hands=s0 AND small=s1 AND yellow=s1	stripes $\in [0.15, 0.40]$; hooves ≥ 0.70; paws < 0.15; hands < 0.15; small $\in [0.15, 0.40]$; yellow $\in [0.15, 0.40]$	antelope	132	0.871
R0004	hunter=s0 AND water=s0 AND small=s0 AND jungle=s2 AND yellow=s1	hunter < 0.15; water < 0.15; small < 0.15; jungle ≥ 0.40; yellow $\in [0.15, 0.40]$	zebra	434	0.866

* Support refers to the number of instances covered in the training partition.

The diagnostic distributions of rule support and confidence, shown in Figure 4, along with WEDD threshold and coverage tradeoff examples depicted, further confirm that the majority of rules are anchored in statistically significant clusters of the semantic space.



(**a**) (**b**) (**c**)

Figure 4. Rulebook and threshold optimization. (**a**) WEDD threshold example on a selected AwA2 semantic attribute. The plot shows how entropy and density jointly determine an interpretable threshold. (**b**) Rule support and confidence distribution for the AwA2 Protocol A rulebook. (**c**) Coverage, abstention, covered accuracy, and covered fidelity as the minimum rule-confidence threshold varies.

By providing these metrics and concrete examples, the framework fulfills the requirement for algorithmic transparency, offering a clear evidence base that moves beyond superficial heatmaps to verifiable logical statements.

4.3. Zero-Shot Transfer as Semantic Validation (Protocol B)

To test whether the transition matrix **T** transfers semantic structure beyond seen training classes, we evaluate the framework under the official xlsa17 split (Protocol B). This experiment is a semantic-transfer validation: the transition matrix is trained only on seen classes and evaluated on unseen classes, but the result is not presented as a competitive zero-shot learning method.

4.3.1. Contextual Performance and Semantic Integrity

The semantic-transfer result is contextualized against established ZSL baselines in Table 10. The SEMTRA continuous prototype variant achieved 44.02% $\pm$ 1.22% unseen-object accuracy across five seeds. Because SEMTRA is optimized for auditability and not for predictive ZSL optimization, the baseline values are included only to orient the difficulty of the split.

Table 10. Contextual quantitative comparison with published AwA2 zero-shot baselines. SEMTRA values are semantic-transfer validation outputs, not leaderboard claims.

Method	Family	Accuracy (%)	Source and scope
DAP [22]	Attribute transfer	46.10	Published ZSL baseline
IAP [23]	Attribute transfer	35.90	Published ZSL baseline
Ours: Semantic Transition	Post-hoc bridge	44.02 $\pm$ 1.22	Object accuracy across five seeds
Ours: Symbolic Template	Post-hoc rule	23.48 $\pm$ 1.70	Object accuracy across five seeds
GFZSL [24]	Generative	63.80	Published ZSL baseline

Specialized generative frameworks like GFZSL achieve higher accuracy (63.80%) because they optimize predictive transfer directly. SEMTRA instead uses Protocol B to audit whether a simple semantic transition can retain class-level structure across seen/unseen boundaries. The symbolic template variant drops to 23.48% ± 1.70%, quantifying the additional information tax incurred when continuous predictions are forced into rigid symbolic templates.

Table 11. AwA2 Protocol B semantic-transfer validation across seeds. This is not a competitive zero-shot leaderboard.

Metric	Mean ± SD
Prototype unseen accuracy	0.4402 ± 0.0122
Symbolic-template unseen accuracy	0.2348 ± 0.0170
Unseen semantic MAE	0.1888 ± 0.0004

4.3.2. Diagnostic Per-Class Analysis

A deeper audit of the seed-42 transfer performance, detailed in Table 12, reveals taxonomic strengths and blind spots. The transition matrix recovered strong visual-semantic structure for blue whale, bobcat, sheep, and rat, while giraffe, dolphin, and bat remained weak under the continuous prototype metric.

Table 12. Official AwA2 xlsa17 unseen-class transfer results (SEMTRA).

Unseen class	Prototype acc.	Symbolic template acc.	Mean Hamming
blue whale	0.9885	0.8046	0.2008
bobcat	0.9333	0.7460	0.4608
sheep	0.8493	0.0437	0.3766
rat	0.8806	0.1032	0.4027
horse	0.4815	0.4857	0.3387
walrus	0.4930	0.1163	0.3930
seal	0.3532	0.1498	0.3753
bat	0.0078	0.0548	0.3829
giraffe	0.0100	0.0241	0.4607
dolphin	0.0254	0.0201	0.2862
Average across five seeds	0.4402	0.2348	0.3714

Conversely, classes such as bat, giraffe, and dolphin exhibit severe semantic-transfer failures, and sheep shows a substantial symbolic-template collapse despite a strong continuous-prototype score. For an auditor, these failures are useful diagnostic signals: they indicate where the ResNet features and the class-level attribute dictionary do not provide the information required for a stable symbolic template. This rupture could be mitigated in future iterations by expanding the attribute dictionary $\mathcal{C}$ to include more granular descriptors or by synthesizing task-specific conceptual variables, but those mitigations are outside the present evidence. This level of class-wise transparency is illustrated in Figure 5, which should be read as a conceptual comparison rather than as a quantitative result.

Protocol B therefore validates the transition matrix as a semantic-transfer diagnostic. It also exposes where symbolic transfer fails, which is central to SEMTRA's audit purpose.

4.4. Cross-Domain Technical Validation

The v1 revision extends the audit protocol beyond AwA2 using SUN and Derm7pt. Table 13 summarizes the generated outputs. The SUN row uses the xlsa17 SUN release with 102 scene attributes; the official SUN Attribute Database paper reports 14,340 images and 707 scene categories, while the xlsa17 release exposes 717 class entries. The Derm7pt row uses dermoscopic images, official case-level splits, grouped diagnosis labels, and seven checklist concepts. Derm7pt is reported only as retrospective technical validation.

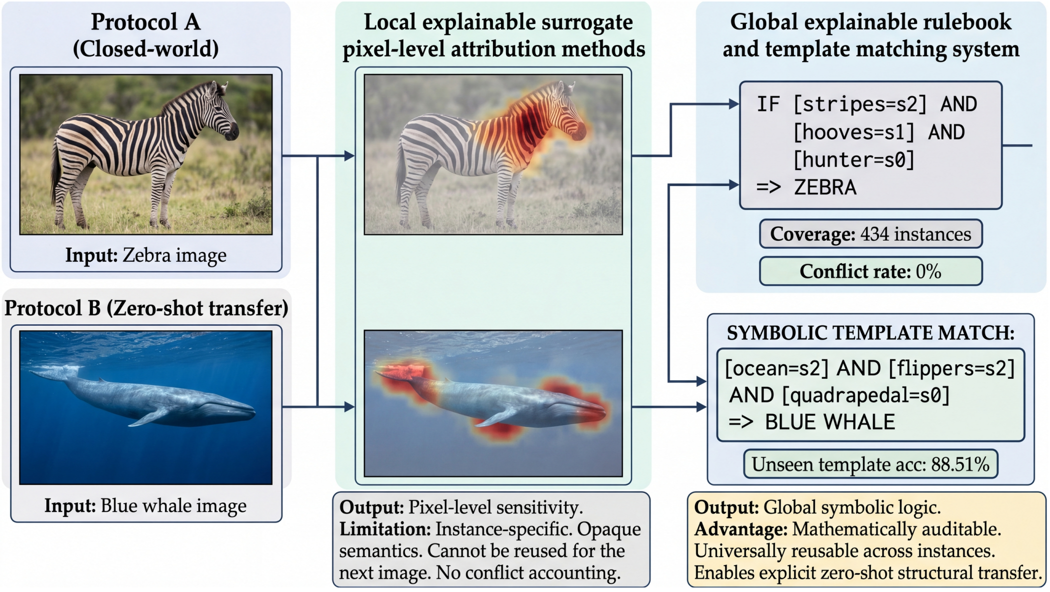


Figure 5. Conceptual comparison between local explainable surrogate pixel-level attribution methods and the proposed global explainable rulebook. The figure is illustrative: it contrasts explanation object types and conflict accounting, not measured predictive performance.

Table 13. Cross-domain SEMTRA revision-v1 summary. Derm7pt is retrospective technical validation only.

Dataset	Scope	Attrs	MAE	Cov.	Fid.	Acc.
AwA2	Protocol A closed-world audit	85	0.1029	0.8480	0.4048	0.3973
AwA2 Protocol B	official xlsa17 semantic-transfer validation	85	0.1888	–	–	0.4402
SUN	xlsa17 SUN unseen transfer and seen-test audit	102	0.0683	0.0760	0.0918	0.0714
Derm7pt	official test split; retrospective technical validation	7	0.2433	0.9241	0.5288	0.4219

The SUN rulebook audit shows low coverage and low covered fidelity, indicating that the current global symbolic template is weak for fine-grained scene categories under this simple feature protocol. Derm7pt shows high coverage but moderate covered fidelity and covered accuracy, which is useful for technical stress testing but insufficient for clinical claims.

4.5. Comparison with Local Surrogates and Concept Baselines

To contextualize the scientific value of the SEMTRA rulebook, we compare its performance against two dominant paradigms in XAI: local post-hoc attributions (LIME/SHAP) and interpretable-by-design concept models (CBMs).

4.5.1. The Fragmentation Gap: SEMTRA vs. Local Surrogates

Local methods like LIME and SHAP provide instance-specific sensitivity rankings, typically visualized as heatmaps. However, as shown in Table 14, there is only a moderate agreement (0.2610–0.3213) between these local attributions and the global antecedents extracted by SEMTRA.

Table 14. Agreement between local post-hoc explanations and fired global-rule antecedents.

Method	n	Top-k agreement	Direction agreement	Rank correlation	95% CI
LIME-style	1000	0.2610	0.5907	0.0222	[0.24, 0.27]
SHAP-style	1000	0.3213	0.6071	0.1271	[0.30, 0.33]

This divergence highlights the fragmentation gap: local methods are highly sensitive to instance-specific noise and pixel-level textures, whereas SEMTRA identifies the stable, global logical structures used by the model across the entire dataset. For an auditor, SEMTRA's rulebook provides a consistent evidence base that local surrogates cannot replicate.

4.5.2. Post-hoc Auditability vs. Ante-hoc Design

We further compared SEMTRA against CBMs, which are specifically trained to predict concepts before labels. As summarized in Table 15, SEMTRA's post-hoc reconstruction error (MAE 0.1076) is virtually identical to that of a dedicated frozen-feature CBM.

Table 15. Consolidated comparison of SEMTRA against concept-based baselines.

Component	Type	Accuracy / Concept Relevance	MAE	Concept Corr.
Frozen-feature CBM	Ante-hoc (Trained)	0.7232	0.1076	0.7759
SEMTRA (Ours)	Post-hoc (Audit)	0.4073	0.1076	0.7759
TCAV	Concept direction	0.4850	–	–

While the CBM achieves higher predictive accuracy because it is structurally optimized for a specific concept vocabulary, SEMTRA provides a comparable level of semantic reconstruction integrity without requiring any retraining of the underlying black-box. The result supports SEMTRA as a practical post-hoc audit layer for frozen representations when an appropriate concept dictionary is available, rather than as a universal substitute for interpretable-by-design architectures.

Unlike traditional symbolic learners that provide a single path to a decision, SEMTRA's use of rough-set granulation allows for a diagnostic audit. While a CART decision tree forces a prediction (leading to zero abstentions but potentially high error in boundary regions), SEMTRA identifies where the model's logic becomes indiscernible. This makes SEMTRA not just a classifier, but a diagnostic tool for identifying the limits of a neural network's conceptual understanding.

4.6. Algorithmic Recovery (Synthetic Benchmark)

To isolate the framework's internal logic from the inherent noise and ambiguity of real-world neural representations, we conducted a comprehensive evaluation using a controlled synthetic benchmark. This stage serves as the algorithmic gold standard, proving that the combination of WEDD discretization and rough-set induction can accurately recover a known ground-truth logic when the semantic signal is clearly defined.

4.6.1. Exactness of Logic Extraction

The synthetic environment consisted of a 10-dimensional feature space with a rule dictionary of known logical structures (Table 16).

Table 16. Synthetic benchmark ground-truth rule dictionary.

Class	Ground-truth rule	Numeric cut summary
1	b_1 high AND b_3 low	$b_1 > 0.70$; $b_3 < 0.35$
2	b_2 medium AND b_5 high	$0.35 < b_2 \leq 0.65$; $b_5 > 0.70$
3	b_4 low, b_6 high, AND b_8 medium	$b_4 < 0.35$; $b_6 > 0.70$; $0.35 < b_8 \leq 0.65$

As summarized in Table 17, at zero noise ($\sigma = 0.000$), the framework achieved a macro-F1 score of 0.879 and a rule Jaccard similarity of 0.700, with a near-perfect operational coverage of 97.10%. The minor baseline threshold error (0.014) observed at zero noise stems from the Gaussian Kernel Density Estimation (KDE) bandwidth smoothing used in the WEDD algorithm, rather than an extraction failure.

Table 17. Synthetic stress-test results averaged over random seeds.

Noise (σ)	MAE	Threshold error	Rule Jaccard	Macro-F1	Coverage
0.000	0.000	0.014	0.700	0.879	0.971
0.100	0.021	0.018	0.763	0.881	0.962
0.200	0.041	0.017	0.743	0.838	0.952

4.6.2. Robustness Under Controlled Noise

The stability of the framework was tested by injecting increasing levels of semantic noise ($\sigma \in [0, 0.20]$). As visualized in Figure 6, the operational coverage remained decisively stable even as noise increased, confirming that the WEDD algorithm successfully anchors thresholds in stable regions.

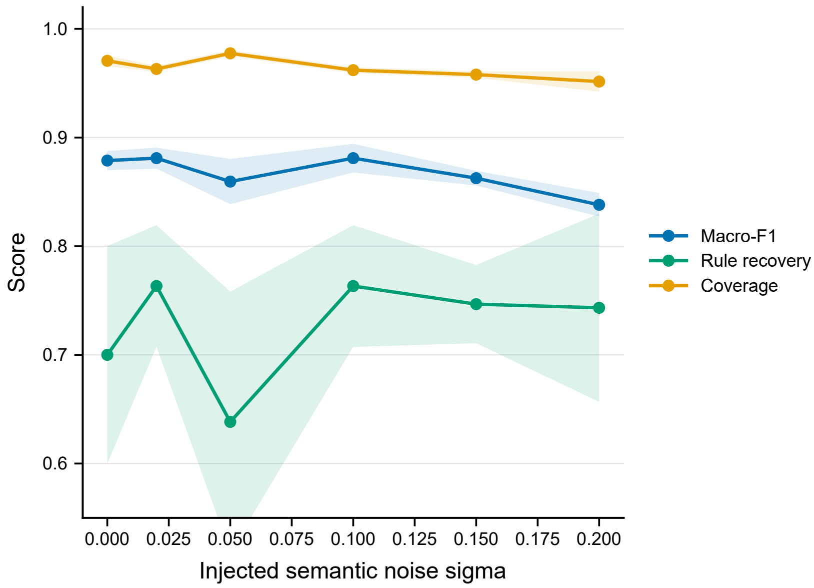


Figure 6. Synthetic benchmark with 95% confidence bands across seeds. Macro-F1, rule-recovery Jaccard, and coverage remain high across the tested semantic-noise range.

While the exact threshold recovery error inevitably fluctuates (Figure 7), the framework maintained a macro-F1 score above 0.838 throughout the tested noise range.

The synthetic results demonstrate that the core mathematical machinery of SEMTRA, specifically the ability to translate continuous variables into discrete granules and extract minimal reducts, behaves predictably when the semantic signal is controlled. For an auditor, this is a useful finding: it suggests that the lower fidelity observed in the real-world AwA2 experiments is driven at least partly by semantic indiscernibility in the frozen features and class-level attributes, rather than by an isolated failure of the rule-induction routine. Consequently, SEMTRA is best viewed as a diagnostic audit engine that reports the verifiable portion of a model's semantic logic, including the portions where that logic fragments or becomes noisy.

Overall, SEMTRA provides a reproducible pathway for deep-learning verification by shifting the explanation artifact from local attribution maps toward global, inspectable symbolic rules with explicit coverage and uncertainty accounting.

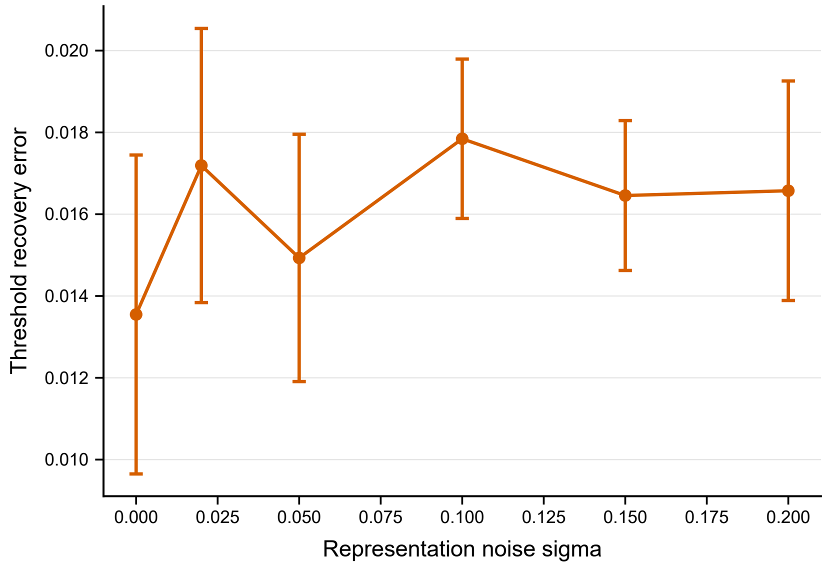


Figure 7. Synthetic threshold recovery error across noise levels.

5. Discussion

5.1. The Scientific Value of Global Auditability

The primary objective of this study was to establish a reproducible, post-hoc pathway that maps semantically opaque neural representations into auditable symbolic knowledge. The v1 evidence shows that this semantic gap can be partially bridged through a global linear transition matrix and rough-set granulation, provided that the concept dictionary is meaningful for the dataset. This addresses the need for verifiable symbolic knowledge identified in recent XAI evolution manifestos [1] while preserving an explicit account of coverage limits.

The current landscape of post-hoc explainability is dominated by instance-specific local methods, such as LIME and SHAP, which generate fragmented attribution heatmaps rather than unified policies [8]. As demonstrated in our comparative results (Section 4.5), there is only moderate agreement between these local feature attributions and the global antecedents extracted by our proposed framework (Table 14). This divergence highlights the fragmentation gap: local surrogate models can be sensitive to instance-specific noise and do not by themselves produce a dataset-level logic policy for auditing.

In contrast, SEMTRA leverages a global transition matrix to identify dataset-wide semantic signals. By shifting the output format to a rulebook with explicit coverage, fidelity, conflict, and abstention reporting (Equation (1)), the framework provides human auditors with a reusable logical artifact. For high-stakes domains, this transition from localized visual approximation to global logical auditing is useful only when paired with careful validation and domain-specific safeguards.

5.2. Interpreting the Audit Tax: Accuracy vs. Verifiability

A critical finding of this research is the quantitative reduction from the base neural predictor's 71.16% accuracy (Table A2) to the rulebook's 39.73% $\pm$ 1.09% covered accuracy and 40.48% $\pm$ 0.78% covered fidelity on non-abstained AwA2 cases (Table 5). We define this reduction as the audit tax: the cost of enforcing discretization (Equation (12)) and logical consistency in a high-dimensional, non-linear latent system. The point is not that the symbolic layer outperforms the base model, but that it exposes which subset of the model's behavior can be reproduced through simple semantic predicates.

The SVD reduction intentionally screens high-frequency representational noise, simultaneously discarding some feature information. The v1 rank-sensitivity grid shows that increasing the retained rank can change the coverage–fidelity balance, but it also increases downstream complexity and runtime. The base model's 71.16% accuracy is therefore a useful predictive reference point, while SEMTRA reports the portion of that behavior that survives the added audit constraints.

Conversely, SEMTRA's covered accuracy and covered fidelity quantify the subset of the neural network's behavior that is currently reproducible through simple semantic predicates. As shown in Figure 3, the framework deliberately operates in a high-coverage regime while accounting for structural uncertainty. The synthetic benchmark evaluations (Section 4.6) show that the rough-set machinery behaves consistently under controlled semantic signal (Table 17). The audit tax observed on AwA2 is therefore consistent with semantic indiscernibility in the ResNet features and class-level attributes, although it should not be interpreted as a complete explanation of every base-model error.

5.3. The Power of Honest Silence and Indiscernibility

The integration of formal rough-set theory [6,7] into the SEMTRA pipeline introduces explicit uncertainty accounting that is often absent from traditional symbolic rule learners like Classification and Regression Trees (CART) [15]. Standard heuristic models force a deterministic classification even when the underlying data manifold is ambiguous. In contrast, the v1 AwA2 rulebook reports a mean conflict rate of $13.75\% \pm 5.32\%$ (Equation (23)) and a mean abstention rate of $15.20\% \pm 5.30\%$ (Equation (22)).

By computing precise lower approximations (Equation (14)), the framework deliberately utilizes structural abstention to transparently signal whenever the underlying neural representation becomes semantically compromised or entirely indiscernible. This honest silence exposes a fundamental property of modern deep feature extractors: they often conflate visually similar but taxonomically distinct classes into identical, inseparable latent regions. When the base predictor cannot semantically distinguish between two distinct concepts, the resulting rough-set boundary region acts as an essential diagnostic warning.

This structural design turns abstention into an audit signal rather than an implementation failure. It also clarifies the comparison with separate-and-conquer rule induction [16]: SEMTRA achieves higher coverage in the reported AwA2 baseline comparison, but its covered accuracy and fidelity remain limited by the semantic bridge and the selected rule granules.

5.4. Linear Transition as a Defensive Choice

The selection of a purely linear ridge-regression operator (Equation (5)) to map complex, high-dimensional latent manifolds represents a highly deliberate, defensive strategy optimized explicitly for structural stability and extreme algebraic transparency. Our ablation studies (Table 3) empirically justify this specific choice; while complex non-linear regressors, such as MLPs, achieve marginally lower continuous reconstruction errors, they simultaneously introduce severe black-box opacity into the semantic transition bridge itself.

This opacity can degrade downstream symbolic logic, reducing rule coverage and increasing structural conflicts. The utility of structured transition matrices has been reported in visual analytics [3] and healthcare informatics [4]. In SEMTRA, linearity serves as a practical safeguard for auditability because the transition weights and reconstruction errors remain directly inspectable.

Consequently, perturbation diagnostics (Table A5) can be interpreted in the same semantic coordinate system as the learned transition. By choosing an auditable linear operator over less transparent non-linear alternatives, the Neural-to-Semantic transition becomes easier to inspect, reproduce, and stress test before the downstream rulebook is trusted.

5.5. Limitations and Strategic Directions

The v1 revision also clarifies several limitations. First, the framework relies on predefined semantic annotations, so it may miss intra-class visual variation, lighting, viewpoint, or texture cues that are not represented in the concept dictionary. Second, because the transition mapping, WEDD discretization, and rough-set granulation are executed sequentially, reconstruction errors can propagate into rule coverage and fidelity. Third, minimal reduct search remains combinatorial, making large attribute dictionaries expensive without pruning. Fourth, the cross-domain results are dataset-dependent: SUN shows low coverage and low covered fidelity under the current scene-feature protocol, whereas Derm7pt shows high coverage but only moderate fidelity and accuracy. Derm7pt should therefore be interpreted as retrospective technical validation only, not as clinical readiness evidence. Future work should explore instance-level concept supervision, differentiable or search-efficient rough-set variants, and domain-specific validation before SEMTRA is used in operational decision support.

6. Conclusions

In this study, we introduced SEMTRA as a post-hoc audit framework that maps frozen representation features into semantic attributes, discretizes those attributes, and extracts rough-set production rules with explicit coverage, fidelity, conflict, and abstention reporting. In the v1 five-seed AwA2 regeneration, the semantic transition achieved a test MAE of 0.1029 ± 0.0005. The extracted rulebook covered 84.80% of test instances, with covered accuracy of 39.73% and covered fidelity to the base predictor of 40.48%. Under the official xlsa17 Protocol B split, the continuous semantic-prototype transfer reached $44.02\% \pm 1.22\%$ unseen-object accuracy; this value is a semantic-transfer diagnostic, not a competitive zero-shot learning claim. The synthetic benchmark reached macro-F1 $= 0.879$ at zero semantic noise and 0.838 at the highest evaluated noise level, showing predictable behavior under controlled signal rather than exact recovery. Cross-domain SUN and Derm7pt runs further show that SEMTRA is portable as an audit protocol but strongly dataset-dependent.

Future research should address these challenges by adding instance-level semantic supervision, improving search efficiency for large concept dictionaries, and validating domain-specific protocols before drawing operational or clinical conclusions.

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Data Availability Statement: AwA2 is available from the official Animals with Attributes 2 dataset page [21]; xlsa17 AwA2/SUN features are available from the MPI zero-shot benchmark resources [20]; SUN Attribute Database metadata are available from the official SUN Attribute Database page [42]; Derm7pt metadata and official split files are available from the authors' public resources [44,45]. The

GitHub repository (https://github.com/radiukpavlo/transition-matrix-dss) includes scripts and generated revision artifacts under `outputs/revision_v1/`. Raw datasets and downloaded archives are not redistributed.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
AUROC	Area Under the Receiver Operating Characteristic
AwA2	Animals with Attributes 2
CART	Classification and Regression Trees
CBM	Concept Bottleneck Model
Cov	Rulebook Coverage
CPU	Central Processing Unit
CR	Conflict Rate
DAP	Direct Attribute Prediction
ECE	Expected Calibration Error
GFZSL	Generative Framework for Zero-Shot Learning
GPU	Graphics Processing Unit
IAP	Indirect Attribute Prediction
ILSVRC	ImageNet Large Scale Visual Recognition Challenge
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
MAE	Mean Absolute Error
MDLP	Minimum Description Length Principle
MLP	Multi-Layer Perceptron
RBF	Radial Basis Function
ResNet	Residual Network
RMSE	Root Mean Square Error
SEMTRA	Global Semantic Transition
SHAP	Shapley Additive Explanations
SVD	Singular-Value Decomposition
TCAV	Testing with Concept Activation Vectors
WEDD	Weighted Entropy-Density Discretization
XAI	Explainable Artificial Intelligence
ZSL	Zero-Shot Learning

Appendix A. Experimental Protocol and Hyperparameters

The feature extractor utilizes the released ResNet-101 representation natively provided with the AwA2 dataset The base predictor is systematically trained on the representation matrix to provide reproducible probabilities, robust class predictions, and explicit fidelity targets. Table A1 outlines the detailed feature and classifier configurations, while Table A2 reports the corresponding validation and test performance metrics, including the Area Under the Receiver Operating Characteristic (AUROC) and Expected Calibration Error (ECE).

Table A1. Feature extractor, base predictor, and revision-experiment hyperparameters.

Component	**Value**	**Implementation detail**
Feature extractor	ResNet-101 features released with AwA2	ILSVRC-pretrained representation layer; no image-level augmentation in this package
Feature dimension	2048	Global average/penultimate representation coordinates
Auxiliary compression	64	Principal Component Analysis (PCA) reduced to 64 components retaining 95% variance
Base predictor	Ridge classifier	Trained on variance-screened representation coordinates
Random seeds	42, 43, 44, 45, 46	Used for v1 AwA2 seed-wise regeneration
Semantic bridge	Ridge regression	Grid alpha in {0.01, 0.1, 1, 10, 100}
Rule thresholds	WEDD	alpha=0.65, max_depth=2, min_bin_size=30, min_gain=0.002
Rule induction	Greedy reducts	tau=0.84, s_min=18 for AwA2 Protocol A
Soft matching threshold	$\tau_H = 0.25$	Maximum allowed masked Hamming distance

Table A2. Base predictor performance on AwA2 Protocol A.

Split	Top-1	Top-5	Macro-F1	Weighted-F1	AUROC	ECE
Validation	0.7093	0.9306	0.5399	0.6639	0.9832	0.6622
Test	0.7116	0.9291	0.5434	0.6681	0.9836	0.6645

It is noted that the optimal α hit the lower bound of the grid search space (0.01); thus, exploring smaller regularization values could potentially yield marginal improvements. Furthermore, the significant disparity between the Top-1 Accuracy (70.93%) and the Macro-F1 score (53.99%) indicates that the base Ridge classifier experiences partial majority-class collapse, heavily favoring well-represented classes over minority taxonomies due to class imbalance.

Table A3 reports the one-factor sensitivity grid for the two auditor-facing control knobs. The results show that increasing λ_s changes the selected semantic basis and modifies rulebook compactness, while increasing λ_H shifts WEDD toward entropy-dominant thresholds and affects the coverage–fidelity balance.

Table A3. Sensitivity of the auditor control knobs on rulebook compactness, coverage, fidelity, and covered accuracy. The λ_s sweep varies semantic-attribute selection while holding $\lambda_H = 0.65$; the λ_H sweep varies WEDD thresholding using the released default attribute set corresponding to $\lambda_s = 0.50$.

Control knob	Value	Rules	Rulebook Coverage	Covered Fidelity	Covered Accuracy
λ_s	0.25	50	0.7687	0.4796	0.4414
λ_s	0.50	54	0.8640	0.3829	0.4073
λ_s	0.75	57	0.7847	0.3856	0.3935
λ_H	0.25	58	0.7136	0.3514	0.3535
λ_H	0.65	54	0.8640	0.3829	0.4073
λ_H	0.90	53	0.8977	0.4349	0.4205

Appendix B. Full Semantic Attribute Discretization Diagnostics

The v1 revision moves the WEDD-vs-MDLP-like entropy comparison into the main Results section because it directly affects the central discretization claim (Table 6). The full machine-readable seed-wise and paired-statistics CSV artifacts are stored under `outputs/revision_v1/awa2/`. These artifacts preserve the complete numerical basis for the reported conclusion: WEDD is a density-aware, stability-oriented option, not an unconditionally superior discretizer.

Appendix C. Rule Inference Traces and Structural Perturbation Stability

Detailed rule inference traces and structural stability analyses are securely retained in this appendix as essential programmatic audit artifacts. Table A4 illustrates representative inference traces successfully mapping individual objects to explicit rule antecedents.

Table A4. Representative rule inference traces with object ID, prediction, matched rule evidence, and antecedent states.

Case	Object	True	Pred.	Mode	Rule	Support	Conf.	Antecedent states
correct exact	5083	collie	collie	exact	R0027	1675	0.208	stripes=s0; hooves=s0; swims=s1; paws=s3; longneck=s0; hunter=s1; ocean=s1; quadrapedal=s3
abstention	24024	ox	abstain	exact	R0029	2820	0.195	stripes=s0; hooves=s2; swims=s1; paws=s0; longneck=s1; hunter=s0; ocean=s1; quadrapedal=s2
fallback error or boundary	16781	hamster	mole	fallback	–	–	–	stripes=s0; hooves=s1; swims=s1; paws=s2; longneck=s0; hunter=s0; ocean=s1; quadrapedal=s2

Appendix C.1. Trace Walkthrough for a Zebra Instance

For test object 36914 (zebra), the 2048-dimensional ResNet representation $\mathbf{a}_{36914}$ is mapped through the transition matrix $\mathbf{T}$ to obtain the reconstructed continuous semantic vector $\hat{\mathbf{b}}_{36914}$. The relevant WEDD-discretized components are $\hat{b}_{\text{hunter}} = 0.0000$, $\hat{b}_{\text{water}} = 0.0501$, $\hat{b}_{\text{small}} = 0.0000$, $\hat{b}_{\text{jungle}} = 0.2632$, and $\hat{b}_{\text{yellow}} = 0.0879$. Using the learned WEDD thresholds, these values become hunter=s_0, water=s_0, small=s_0, jungle=s_2, and yellow=s_1. This state vector exactly satisfies Rule R0004, whose antecedent is hunter=s_0 AND water=s_0 AND small=s_0 AND jungle=s_2 AND yellow=s_1, yielding a verified zebra prediction with support 434 and confidence 0.866.

Furthermore, Table A5 rigorously reports the operational rule consistency and prediction stability under varying degrees of semantic-representation perturbations. Reconstructed semantic representations were perturbed using additive Gaussian noise:

$$\hat{\mathbf{b}}_{i,\text{perturbed}} = \hat{\mathbf{b}}_i + \boldsymbol{\epsilon}_i, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2 \hat{s}_j^2),$$

where $\hat{s}_j$ is the empirical standard deviation of reconstructed semantic attribute j on the test split.

Table A5. Rule consistency under semantic-representation perturbations.

Sigma	Rule consistency	Decision consistency	Coverage	Coverage change	Abstention	Conflict
0.0000	1.0000	1.0000	0.8796	0.0000	0.1204	0.1180
0.0100	0.9745	0.9663	0.8792	-0.0004	0.1208	0.1183
0.0250	0.9326	0.9176	0.8808	0.0012	0.1192	0.1168
0.0500	0.8667	0.8401	0.8816	0.0020	0.1184	0.1160
0.1000	0.7307	0.7112	0.8940	0.0145	0.1060	0.1026

Because WEDD anchors thresholds in low-density valleys rather than dense class-overlap regions, rules exhibit high robustness against micro-perturbations up to $\sigma = 0.05$. The degradation curve is therefore gradual: rule consistency remains 0.8667 and coverage remains essentially unchanged at 0.8816 at $\sigma = 0.05$, while the larger $\sigma = 0.10$ setting produces the expected decline in rule and decision consistency.

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